# The reciprocal function

## Lecture and worked examples

Maths Explorer

## Reference notes

$f'(x) = -\dfrac{1}{x^2} < 0$ for every $x \neq 0$. The reciprocal function is strictly decreasing on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$. The lines $x = 0$ and $y = 0$ are asymptotes.

## Reference notes

One cannot write that the reciprocal function is decreasing “on $\mathbb{R} \setminus \{0\}$” without specifying the two intervals. For example $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$, which would contradict the global definition of decreasing.

## Reference notes

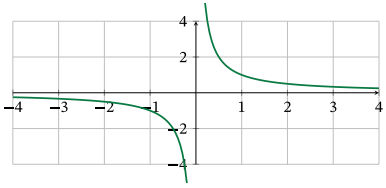
On $]0, +\infty[$, $g(x) = x + \dfrac{4}{x}$ satisfies $g'(x) = 1 - \dfrac{4}{x^2} = \dfrac{(x-2)(x+2)}{x^2}$. The denominator is positive; $g'$ is negative on $]0, 2[$ then positive on $]2, +\infty[$. The function has a minimum $g(2) = 4$.

# Definition and graph of the reciprocal function

- Find the domain.
- Draw the curve.
- Use symmetries.

# Definition and graph of the reciprocal function

Calculate $\frac{1}{x}$ for $x = -4, -2, -1, -\frac{1}{2}, \frac{1}{2}, 1, 2, 4$. Plot the points and describe the symmetries. Why is $x = 0$ missing?



## Definition and graph of the reciprocal function

The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its curve is a hyperbola symmetric about the origin.

## Definition and graph of the reciprocal function

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

## Definition and graph of the reciprocal function

**1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

**2 h session:** add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

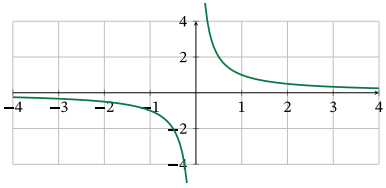
**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

# Behaviour at the boundaries and asymptotes

- Describe the behaviour at the boundaries.
- Identify the asymptotes.
- Read a graph.

# Behaviour at the boundaries and asymptotes

Using a table of values near 0 and for large values of $|x|$, describe $\frac{1}{x}$. Translate each behaviour into an asymptote and a sentence.



## Behaviour at the boundaries and asymptotes

For $f(x) = \dfrac{1}{x}$, $x = 0$ is a vertical asymptote and $y = 0$ a horizontal asymptote. The branches approach them without crossing them.

## Behaviour at the boundaries and asymptotes

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## Derivative and direction of variation

- Differentiate the reciprocal function.
- Justify the variations.
- Draw up a table.

## Derivative and direction of variation

Starting from the difference quotient of $f(x) = \dfrac{1}{x}$, recover $f'(x) = -\dfrac{1}{x^2}$. Study its sign and check consistency with the graph.

## Derivative and direction of variation

For $x \neq 0$, $f'(x) = -\dfrac{1}{x^2} < 0$. The reciprocal function decreases on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$.

## Derivative and direction of variation

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# Combinations with polynomial functions

- Differentiate a combination of functions.
- Simplify an expression.
- Organise a function study.

## Combinations with polynomial functions

A company produces $x > 0$ hundred items and its average cost is $C(x) = 2x + 30 + \dfrac{200}{x}$. Calculate $C'(x)$, factorise its numerator and find the production level that minimises the average cost.

## Combinations with polynomial functions

To study $P(x) + \dfrac{k}{x}$, find the domain, calculate the derivative, put over a common denominator, study its sign, then draw up the variations.

## Combinations with polynomial functions

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## Sign and factorisation

- Factorise.
- Build a sign table.
- Respect excluded values.

## Sign and factorisation

Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Build the table with the three critical values, distinguish a zero from an excluded value, then solve the inequality $\dfrac{(x-2)(x+3)}{x} \geq 0$.

## Sign and factorisation

In a quotient, the sign depends on the numerator and the denominator. Zeros of the denominator remain excluded from the domain, even if a factor formally cancels; they are marked as excluded values in the sign table.

## Sign and factorisation

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# Derivatives of polynomials of degree at most 3

- Differentiate a polynomial.
- Study a derivative.
- Deduce the variations.

## Derivatives of polynomials of degree at most 3

For $f(x) = 2x^3 - 3x^2 - 12x + 5$, calculate $f'(x)$, factorise, study its sign and deduce the variations of $f$.

## Derivatives of polynomials of degree at most 3

$(ax^3 + bx^2 + cx + d)' = 3ax^2 + 2bx + c$. The sign of the derivative gives the direction of variation of the function.

## Derivatives of polynomials of degree at most 3

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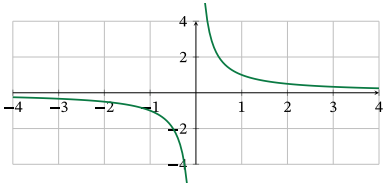
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# Full study and graph

- Carry out a full study.
- Draw a consistent curve.
- Interpret an extremum.

# Full study and graph

Study $g(x) = x + \dfrac{9}{x}$ completely on $]0, +\infty[$: domain, derivative, sign, variations, minimum, behaviour at the boundaries and sketch.



## Full study and graph

A full study includes: domain, derivative, sign, variations, limits or behaviour at the boundaries, key points and a consistent graph.

## Full study and graph

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

## Full study and graph

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## Review and assessment

- Use domain, derivative and sign.
- Write up a study.
- Correct work methodically.

## Review and assessment

Correct a deliberately flawed study of $h(x) = x + \dfrac{4}{x}$: stated domain $\mathbb{R}$, derivative $1 + \dfrac{4}{x^2}$, table with no break at $0$. Rewrite a complete solution.

## Review and assessment

Check the four items separately: domain, expression of the derivative, sign of the derivative, variations. An error at one step must not hide the following ones.

## Review and assessment

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